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Overview of Current GW observations: main techniques and results

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See Figure 1 of the LIGO detector, noise, and data paper: here we'll call this the LIGO noise paper, i.e., LIGONP: https://arxiv.org/abs/1908.11170
Mike explained how the data is obtained, etc. We need to *process* it to search for GWs, then we can do physics with it.

Searches for GWs:

1) The detector is like a giant antenna for spatio-temporal vibrations that are *prepared* in a time series. Generally, this time series is just a bunch of "noises." Aha!... One persons noise is another person's signal! ;)

2) From a GW astronomer's perspective, we can decompose the data *d* into two components: the response of the detector to a GW signal, *h*, and the combination of all the noise sources in the detector, *n*, such that $d = h + n$.

   We do not know a-priori the nature of the signal *h* (though we can have a guess of it's basic form from GR predictions for a compact binary coalescence (CBC)). We need a way to handle this while avoiding statistical pit-falls.
   A basic recipe (C. Berry's abridged guide to LIGONP):
   1. Fourier transform the time series to the frequency domain,
   2. Data quality - make sure we have good data,
   3. Noise model and likelihood function - make sure they're reliable,
   4. Signal detection: multiple approaches with varied strengths/weaknesses,
   5. Inference: waveform and parameter estimation.

3) Data processing with a noisy dataset:
   a. If the *residuals* $r = d - h$ are consistent with our model *h* for the instrument noise, then we consider to be a good description of the signal in the data.
   More quantitatively, the likelihood that the data *d* contains a possible signal *h* is given by the probability that *r* is a realization of the noise model. In other words, the likelihood function is the noise model.

   b. For stationary (i.e., correlations depend only on the time lag between data samples, or the statistics don't change over time) Gaussian noise, the likelihood is, $$p(\boldsymbol{d}|\boldsymbol{h}) = \frac{1}{\det(2\pi \boldsymbol{C})^{1/2}} \mathrm{e}^{-\frac{1}{2}\chi^2(\boldsymbol{d},\mathbf{h})}$$

   where C is the noise correlation matrix and $\chi^2(\boldsymbol{d},\mathbf{h})$ is the chi-squared statistic. Let's translate this to GWs.

   a. For stationary, Gaussian noise that is uncorrelated b/w different detectors, the likelihood is Eq (9) from LIGONP.

   b. *** Note, that LIGO implements methods for measuring noise deviations from stationarity and Gaussianity, see e.g. Sections 8 and 9 of LIGONP. Nevertheless, it is likely that the effects of the non-Gaussian components will be unimportant due to their short duration relative to the length of the signal we're interested in, due to the rejection of events that are not simultaneously detected in two or more detectors, and due to the filtering of the detector outputs with theoretical waveform templates (Cutler&Flanagan1994). ***

   a. The Fourier transform of a function $h(t)$ is, $$\tilde{h}(f) = \int_{-\infty}^{\infty} h(t)\mathrm{e}^{-2\pi i f t}\,\mathrm{d}t$$

   b. For stationary, Gaussian noise, we can use the *power spectral density* (PSD), which is the Fourier transform of the correlation matrix C (see eq. A3 of Cutler&Flanagan1994), to write analytic expressions:

      i. the PSD of (stationary Gaussian) noise, $S_n(f)$, is a diagnoal matrix in the frequency domain.

      ii. Use the Fourier transform to get frequency-domain *d(f)* from time-domain *d(t).*

      iii. But, the integral for Fourier transform is defined over (-inf, inf), but we're interested in a finite time series, so we must *window* our data. That is, we want a function *w(t)* that goes to zero outside of the frequency range we're interested in, such that the Fourier transform of *d(t)w(t)* provides a *convolution* of our data with the window, i.e., *d(f)w(f)*.
      Choosing an appropriate window function is nontrivial and if done incorrectly can lead to unwanted correlations known as spectral leakage (https://en.wikipedia.org/wiki/Spectral_leakage). We always want to use a *tapering window* to avoid distortions of the data and to minimize edge-effects. Note that

the *usual* window used in LSC data analysis is the Welch method (https://en.wikipedia.org/wiki/Welch%27s_method#)

iv. Now with our frequency-domain data, we need to properly weigh the components (like how shouting is less startling at a rock concert than a library), but this is easy: just divide by the PSD:

$$d_w(f) = \frac{d(f)}{[S_n(f)]^{1/2}}$$

This is known *as whitening* the data. It transforms a signal to have equal amplitude fluctuations at all frequencies, i.e., it normalizes the power at all frequencies so that excess power at any frequency is more obvious. This is useful as low-frequency amplitudes can dominate higher frequency amplitudes.

v. For data visualization, you'd want to then bandpass the data which removes features that are outside the frequency band of interest (like seismic noise at low freq, and quantum sensing noise at high freq.). Note that bandpassing is not done in official LVC data analysis.

1. Searching for GWs:
Two kinds of searches: one is quick and dirty, the other is arduous and technical. But, both are sophisticated!

   a. Non-template-based searches: (also called unmodeled searches, or minimally or weakly modeled searches). Give some examples, coherent wave burst (cWB) which was the first to see GW150914 as it searches for compatible signals in multiple detectors assuming the signal increases with time, as is the characteristic "chirp" of a CBC), harmonic component analysis (tries to construct wave-like signals in the data using wavelet approximations/constructions), machine learning, topological data analysis, etc (see https://www.mdpi.com/2076-3417/13/17/9886).
   More about cWB here: https://cplberry.com/tag/neil-cornish/

   b. Template-based searches: (also called "modeled searches", main technique is called "matched filtering analysis". The thesis by Janquart is amazingly in depth for this!), explain the basics of this with the welch window and wigner filter. Explain different types of filters, but this is a detail beyond this lecture.

Typically, a low-latency search is done (meaning small delays b/w data collection, calibrating, etc.) usually with a non-template-based search since these are quick. When that finds a potential signal, it is followed by a template-based search to run better tests for whether it's actually a physical signal and to recover better estimates on the source parameters via Bayesian parameter estimation pipelines. See the image here (https://emfollow.docs.ligo.org/userguide/analysis/index.html) which shows that this process is iterative to ensure we aren't fooling ourselves!

2. The main template-based approach: the *matched-filter analysis* (see e.g., Sathyaprakash and Schutz, 2009):

   a. The probability that there is a signal in our data can be determined by considering the null hypothesis (the data only contains noise) and the signal hypothesis (the data contains noise and a GW signal).

   b. For stationary Gaussian noise, the (log) likelihood becomes,

$$\log(P|d) \sim \int_{-\infty}^{\infty} \frac{|h(f) - d(f)|^2}{S_n(f)} df$$

   which is just the usual noise-weighted inner product (eq. 8 of LIGONP) for a given signal model *h* (like the waveforms from lecture 1).

   c. Apply this likelihood to Bayes' Theorem to compute the *likelihood ratio* (eq. 12 of LIGONP), which if significantly larger than 1 indicates the presence of a GW signal of form given by *h* in the data.

   d. A GW with strain h_{ab} that is incident on a GW detector induces a linear response *h(t)* that is fully characterized by the two response functions $F_{+,x}$, i.e., $h(t) = F_+h_+(t) + F_xh_x(t)$ where h_+ and h_+ are the two polarizations (for $F_{+,x}$, see e.g.'s, CutlerFlanagan1994 for LIGO-type detector and Babak2015 for LISA-type detector).

   e. Let's quantify the difference b/w the detector's responses $h(t)$ and $h_0(t)$ to the source and the template, respectively. That is, we want to compare how the detector responds to a physical signal and how it responds to a model (template) of that signal. If the response is very similar, then we have confidence that the detector is responding to a physical signal! Use the inner product:

$$\langle h_0|h\rangle \equiv 4\,Re \int_0^{\infty} \frac{h_0(f)\,h^*(f)}{S_n(f)} df$$

   and recall that $S_n(f)$ contains the noise model for the detector.

$$\langle h_0|h\rangle = 4\Re\int_0^{\infty} \frac{\tilde{h}_0^*(f)\,\tilde{h}(f)}{S_n(f)}\,df$$

and recall that $S_n(f)$ contains the noise model for the detector.

The overlap of the normalized inner product of the two responses (CutlerFlanagan1994),

$$O(h_0, h) \sim \frac{\langle h_0|h\rangle}{\sqrt{\langle h_0|h_0\rangle\langle h|h\rangle}}$$

The signal-to-noise ratio (SNR) of the source of the physical signal is $\rho^2 \equiv \langle h_0|h_0\rangle$.
In the literature, this is also called the "optimal" SNR of the signal.
It can be computed via,

$$\rho^2 = 4\int_{-\infty}^{\infty} \frac{|h_0(f)|^2}{S_n(f)}\,\mathrm{d}f$$

Two waveforms are distinguishable when their difference is sufficiently large, i.e., $O(h - h_0,\ h - h_0) \gtrsim 1$, and equivalently when the condition is satisfied that $O(h_0, h) \lesssim 1 - \rho^{-2}$.
See eq 29 of LIGONP for a related concept known as the *mismatch.*
In practice, we have *template banks* which are large families of waveform models/approximants that are run through the matched-filter analysis using Bayes' theorem.

A useful quantity is the characteristic strain of a signal: $[h_c(f)]^2 = 4f^2|h_0(f)|^2$
Analogously, we have for the noise of a detector, $[h_n(f)]^2 = fS_n(f)$
This allows us to easily estimate the SNR of a signal by eye on a plot of $h_c(f)$ vs *f.*
The above integral becomes (see Moore, Cole, Berry 2015 for details),

$$\rho^2 = \int_{-\infty}^{\infty} \left[\frac{h_c(f)}{h_n(f)}\right]^2 \mathrm{d}(\log f)$$

6) Other search/detection statistics: false alarm rate (which is the expected rate of events per time due to noise which would be assigned an equal or larger ranking statistic than the candidate), pastro (which encodes the probability that the detected signal is due to an astrophysical source by comparing the differential rates of gravitational waves and background events), etc… See the thesis by Janquart for details about FAR; see Lynch2018 (https://arxiv.org/abs/1803.02880) for details about Pastro. This is an open area of research.

7) For a walkthrough-example of this process with GW150914, see https://cplberry.com/tag/neil-cornish/

8) A note on inference of source parameters:
   "For CBC events, we calculate a number of quantities that are inferred from the signal. In preliminary alerts, these quantities are based on the candidate significance and the matched-filter estimates of the source parameters. Once parameter estimation has been completed, updated values will be provided based on samples drawn from the posterior probability distribution"
   (https://emfollow.docs.ligo.org/userguide/analysis/inference.html)

Parameter Estimation:

(taken from Thrane & Talbot (2019), https://arxiv.org/abs/1809.02293)
(also see LIGONP section 9)

Note: now that we have cleaned up the data and we found a GW signal, let's estimate the parameters of the source of the GW signal.

1) Bayesian inference is the main way of handling probabilistic uncertainty in our measurements to deduce the properties of the source. This works out well because GR gives us extremely precise predictions for the possible source properties.

2) Basic Bayesian set-up for single source:
   a. Our primary goal is to construct the posterior distribution $p(\theta|d)$, where θ is the set of model parameters (ie the 15ish parameters that describe a binary black hole: 2 masses, 2 spin magnitudes, GW phase, inclination, distance, sky location (longitude and latitude), time of coalescence, two angles to describe the

direction of **L**, etc…)

b. We use Bayes' theorem to compute the posterior distributions:

$$p(\theta|d) = \frac{\mathcal{L}(\mathrm{d}|\theta)\pi(\theta)}{Z}$$

where $\mathcal{L}(\mathrm{d}|\theta)$ is the likelihood function, $\pi(\theta)$ is the set of prior distributions (one for each parameter), and the normalization factor $Z = \int \mathcal{L}(\mathrm{d}|\theta)\pi(\theta)d\theta$ is called the *evidence.*

c. Bayes' theorem is about using the data (and perhaps a model) to update the priors to arrive at the posteriors. We choose the likelihood function to be stationary and Gaussian, as earlier.

d. In practice, computing posteriors is very difficult, because there are ~10^15 combinations of parameters that define the likelihood space. So, we often use approximation methods, such as Markov-chain Monte Carlo, nested sampling algorithms, or machine learning approaches which are popular now.
For a recent review, see Christensen & Meyer 2022.

e. **Bayes factor**: the ratio of evidence for two different models. See section 3 of Thrane & Talbot 2019. For examples, we can compare the evidence for model that has no signal (ie, noise only) with the evidence for a model that has signal and noise (as we discussed above); we can compare the evidence for a model that includes spins in the waveform model with the evidence for a model that does not include spins in the waveform; we can compare the evidence for a binary black hole waveform predicted by general relativity with a binary black hole waveform predicted by some other theory.

f. For further discussion about this, see the LIGONP eqtns 27 and 28.

g. This framework allows us to estimate the parameters of individual CBCs. Since the likelihood space is nontrivial, it takes a lot of CPU hours to accomplish this for even a single source.

3) The extension to multiple sources is known as hierarchical Bayesian analysis which uses hyperparameters to model the population where each member of the population has its own parameters as above. This allows us to quantify *population* properties of a set of GW events. See section 5 of Thrane & Talbot 2019.

Main results in the first 3 runs of LIGO.

1) Three GW transients catalogues produced so far:
   a. O1/O2 catalogue papers: https://www.ligo.org/detections/O1O2catalog.php
   b. O3a catalogue papers: https://www.ligo.org/detections/O3acatalog.php
   c. O3b catalogue papers: https://www.ligo.org/detections/O3bcatalog.php
   d. O3 catalogue summary: https://www.ligo.org/science/Publication-O3bCatalog/
   e. Data releases: https://www.ligo.org/scientists/data-releases.php
   f. Zenodo - folded data for first three obs runs: https://zenodo.org/record/6326656
   g. Zenodo - data for O3b: https://zenodo.org/record/5636796

2) Observing run timelines: https://observing.docs.ligo.org/plan/

3) The LVK's three CBC-population data analysis papers:
   a. GWTC-1 population paper
   b. GWTC-2 population paper
   c. GWTC-3 population paper
   Examine figures blah, blah, and blah, of ligopaper

4) Let's look at some plots from the population paper of GWTC-3 (the third GW transient catalogue):
   - Table 1, and Figures 1, 2, 7, 10, 15, 16, 17

Some thoughts:

1) When will we measure relativistic spin precession in the black-hole binaries (BBHs)?
   https://arxiv.org/abs/1908.00555
   Can we measure the individual black-hole spins from GWs?
   https://arxiv.org/abs/1512.04955

The currently low SNR of the LVK detectors (ie $\lesssim 30$) generally doesn't allow us to probe spin precession, see e.g.'s
https://arxiv.org/abs/2111.10455
https://arxiv.org/abs/2204.00026
The below authors propose using their "precession SNR" and Bayes factors and the recovered posteriors to accurately predict if a GW signal has spin precession effects,
https://arxiv.org/abs/2010.04131

2) The event GW200129 is the first event that has been argued to have a clear identification of spin precession, see https://arxiv.org/abs/2112.11300
who show the primary BH spin magnitude is likely $\gtrsim 0.4$ and directed near the orbital plane.
The reported SNR of the event in the LIGO O3b catalogue is 26.5.
Another group responded,
https://arxiv.org/abs/2206.11932
explaining how a glitch occurs at the same time in the datasets, and they do not disprove the existence of spin precession for this source but they stress that careful mitigation of things like glitches is crucial for observing physical effects in GW data.

3) How do we explain the *apparent* correlation b/w aligned effective spin χ_{eff} and mass ratio q ? See, e.g., Figure 21 of GWTC-3 population paper.
Astrophysics is complicated… need more detected sources, at least some with higher SNRs, and maybe some luck :)

A few resources:

The LIGO data analysis and noise paper (LIGONP): https://arxiv.org/abs/1908.11170

C. Berry's abridged guide to the LIGONP:
https://cplberry.com/2020/02/09/gw-data-guides/#data-analysis-guide-short

LIGO docs for online search and data analysis pipelines:
https://emfollow.docs.ligo.org/userguide/analysis/searches.html

Thesis by Janquart about matched filtering statistics theory:
https://matheo.uliege.be/bitstream/2268.2/9211/4/Master_thesis_Janquart.pdf

Example code for handling LIGO public alert notices:
https://emfollow.docs.ligo.org/userguide/tutorial/index.html

Bayesian vs Frequentist statistics:
https://towardsdatascience.com/statistics-are-you-bayesian-or-frequentist-4943f953f21b

Some essential papers:

E. Poisson - parameter estimation of CBCs (1995): https://arxiv.org/abs/gr-qc/9502040

Basics of population rates (2014): https://arxiv.org/abs/1302.5341

Thrane & Talbot - intro to Bayesian inference (2019): https://arxiv.org/abs/1809.02293